ii. The new node then asks successor (r) for its predecessor and then asks both of these to insert r in between them in the circle.

Chord algorithm has been used to construct a distributed file system and other applications.

3.3 Congestion Control Algorithms

Congestion is a situation where too many packets are dumped into subnet by the hosts. Due to this, performance degrades. Congestion in a network occurs if users send data into the subnet at a rate greater than allowed by network resources. Network resources can be communication links, routers, buffers in switches and so on.

Congestion will occur due to many reasons. If the traffic increases, the router are no longer able to cope up and start losing packets. At very high traffic, performance degrades completely and delivery of packets will be less.

Congestion can be brought down by following measures :

i. If there is insufficient memory to hold all the packets in router, packet will be lost. By increasing memory, loss of packets can be controlled.

ii. Slow processor on router can also cause congestion. When CPU performs tasks very slowly, then congestion may occur. Tasks can be queueing buffers, updating tables etc., Adding high speed processor will solve this problem.

iii. Low bandwidth lines can also cause congestion. Upgrading the lines may help to avoid congestion to some extent.

3.3.1 General Principles of Congestion Control

The complex system has various problems, such as computer network. All solutions to these problems are grouped into *(i) open loop (ii) closed loop.*

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Differences between open loop and closed loop

Open Loop	Closed Loop
i. Solution from open loop attempt to solve problem by good design.	i. Solution from closed loop are based on the concept of feed back loop.
ii. Once the system starts running, it is not interrupted in middle to make corrections.	ii. System monitors when and where congestion occurs. Passes the information to places where action can be taken. The system is adjusted to correct the problem.
iii. Several tools are used for open loop controling. Tools decide when to accept new traffic, or when to discard packets etc.,	iii. Monitoring the subnet can be done by using various metrics. The metrics can be lack of buffer space, average queue length, average packet delay etc.,

In above approaches, the knowledge of congestion will cause the hosts to take appropriate action to reduce congestion. Many congestion control algorithms were developed. *Yang and Reddy* have developed a taxonomy for congestion control algorithm. They divided all algorithms into *open loop* and *closed loop*.

Open loop algorithm into those acting at the source versus those that act at the destination.

Close loop algorithms into *explicit feed back* and *implicit feed back*.

In *Explicit feed back algorithms*, packets are sent back from the point of congestion to warn the source node. In *Implicit feed back algorithms*, source node itself confirms congestion by making local observations.

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3.3.2 Congestion Prevention Policies

The congestion prevention policies are discussed at different levels in open loop system. There are different policies that can affect congestion at data link, network and transport policies.

i. Data link layer policies that effects congestion. Data link layer is concerned with four policies
   i. *Retransmission policy*
   ii. *Out of order caching policy*
   iii. *Acknowledgement policy*
   iv. *Flow control policy*

In retransmission policy, when sender waits for acknowledgement, but it fails to get acknowledgement from the other end and sender time out has occured and then once again the packets are to be retransmitted. The retransmission of packets depends on the ARQ protocols used. Go back n will put load on the system, because all the packets from failed one has to be retransmitted again. This leads to congestion heavily. With respect to congestion control, selective repeat is better than go back n if the receivers routinely discard all out of order packets, these packets will have to be transmitted again later, this creates extra load.

Sending acknowledgement for each packet also affects congestion. These acknowledgement packets generate extra traffic. In flow control mechanism, if the window size is big, then there may be chances of losing the packets. Using tight flow control with small window reduces the data rate and fights congestion

ii. Network layer adopts the following policies
   i. *Virtual circuits versus datagram inside the subnet*
   ii. *Packet queueing and service policy*
   iii. *Packet discard policy*
   iv. *Routing algorithm*
   v. *Packet lifetime management*

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In network layer, many congestion control algorithms works only with virtual circuit subnet. So, choosing virtual circuit or datagram subnet affects congestion. Packet queueing and service policy relates to whether routers have one queue per input line, one queue per output line, or both. It also relates to the order in which packets are processed. Packet discard policy, tells which packet is dropped when there is no space. Good policy will *avoid congestion* and bad will make *congestion worse*.

Routing algorithms also play a very important role in avoiding congestion by spreading the traffic uniformly (to some extent) over all the lines. Packet life time must be specified correctly. This deals with how long a packet may live before it is being discarded. If the packet life time is more, it may block up lines for long time. If the packet life time is short, it may die (time out) before reaching destination. Thus, this packets needs retransmission. So, in this way packet life time management policies affect management.

In the transport layer, the same issue occurs as in the data link layer, but in addition, determining the time out interval is difficult. Because the transit time over network is less predictable than the transmit time over a wire between two routers.

If the time out interval is too short, extra packet will be unnecessarily sent due to retransmission. If the time out interval is too long, congestion will reduced, but the response time will suffer whenever packets are lost.

3.3.3 Congestion Control in Virtual Circuit Subnets

In this section congestion control in virtual circuit subnets will be discussed. There are various approaches to control congestion in virtual circuit subnet. Some of these approaches are

i. *Admission control*
ii. *Route all new virtual circuits around problem areas*
iii. *Negotiate host and subnet during virtual circuit set-up*

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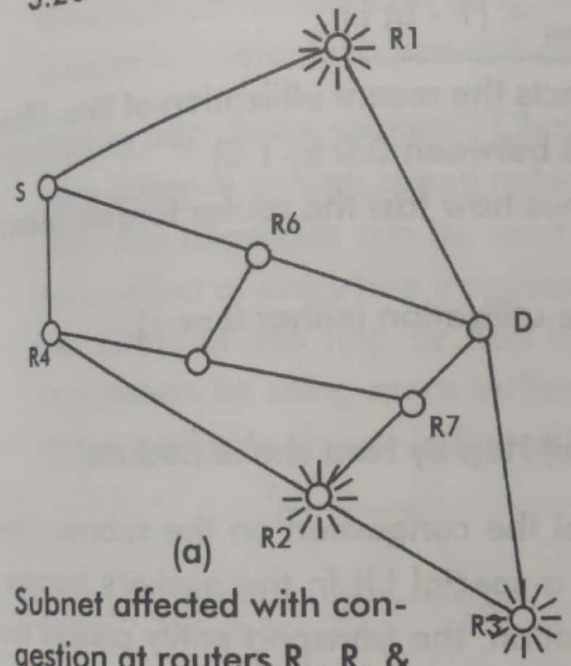
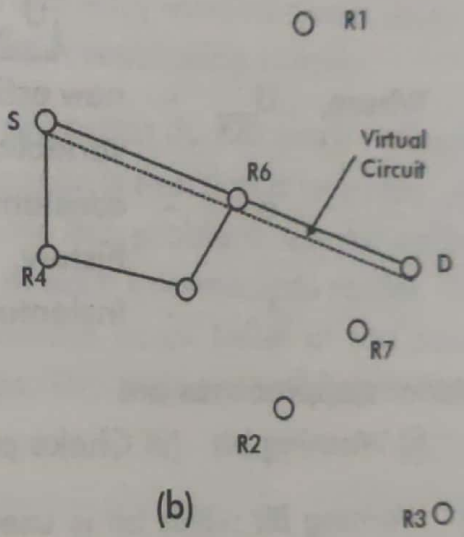
i) *Admission Control* : This is controlled during the new virtual circuit set-up. If there is an indication of congestion (heavy traffic) on the subnet, no new virtual circuit are set-up intil congestion reduces. Thus, attempts to setup new connection at transport layer fails.

ii) *Route all new virtual circuit around new problem areas* : In this approach new connections are allowed but carefully route all new virtual circuit around problem areas. To illustrate this situation, consider the subnet of figure 3.20 in which three router are in congestion.



(a)

Subnet affected with congestion at routers R_1, R_2 & R_3



(b)

New virtual circuit connection eliminates congestion VC from S to D is shown

Fig.3.20(a) Congested subnet (b) A redrawn subnet that eliminate the congestion

Suppose router S wants to setup a connection to another router D. A new route and subnet is considered by omitting the congested routers R1, R2 and R3 and all of their lines. The new route is shown in dashed line, avoids the congested routers.

iii) *Negotiate host and subnet during virtual circuit setup* : The agreement between the host and subnet specifies the volume and shape of the traffic, quality of service required and some other parameters. As a part of the agreement, the subnet will reserve resources during circuit setup. Resources can be table and buffer space in routers and bandwidth on the

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communication lines. In this way, congestion can reduced on the new virtual circuit because all the necessary resources are reserved and available.

3.3.4 Congestion Control in Datagram Subnets

In the section, various approaches that can be used in datagram subnets are discussed. Each router can easily monitor the utilization of its output lines and other resources. The relationship between recent line utilization and old line utilisation is given by the equation.

$$U_{new} = a \cdot u_{old} + (1 - a) f$$

Where, U_{new} - new estimate reflects the recent utilization of line. (Real variable value lies between 0.0 to 1.0)

a - constant determines how fast the router forgets recent history.

f - instantaneous line utilization (either 0 or 1)

Various approaches are

(i) Warning bit (ii) Choke packets (iii) Hop by Hop choke packets

i) *Warning Bit* : This bit is used to signal the congestion on the subnet. The warning state is indicated by setting a special bit in the packet's header. When the packet arrived at its destination, the transport entity copied the bit into the next acknowledgement back to the source. The acknowledgement is sent back to the source indicating the congestion. So, source then reduces the traffic. This warning bit is set as the router is in warning state. As long as warning bits continue to flow in, the source continue to decrease its transmission rate. If warning bit is not set, then transmission rate can be increased.

ii) *Choke Packets* : In this mechanism, special choke packets are used to inform about the congestion back to source machine. The router sends choke packet back to source, giving information about congestion on that line or at the destination. If there is no congestion, then original packet is tagged so that it will not generate any more choke packets further along the path. Packets are then forwarded in the usual way.

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When the source machine gets the choke packet, it is necessary to reduce the traffic sent to the specified destination by some percent. At every fixed interval of time, host will try to listen to choke packets. If choke packets arrives at that time interval, this indicates line is still congested, so the source machine reduces the flow still more and ignore choke packets for fixed time interval. If there is no choke packets during the listening period, then source machine may increase the flow again.

Source machine can reduce traffic by adjusting their of policy parameters. One such parameter can be *window size*. Increasing window size is done in smaller increments to prevent congestion from reoccuring quickly.

iii) *Hop by hop choke packets* : Choke packet approach do not work well with long distance or high speed network. The reason is reaction is *very slow*. An alternate approach can be used to overcome this problem. Choke packet takes effect at every hop it passes through, at each intermediate router, The net effect of this hop by hop scheme provides quick relief at the point congestion by using more buffers upstream. The sequence of hop by hop choke packet is shown in figure 3.21.

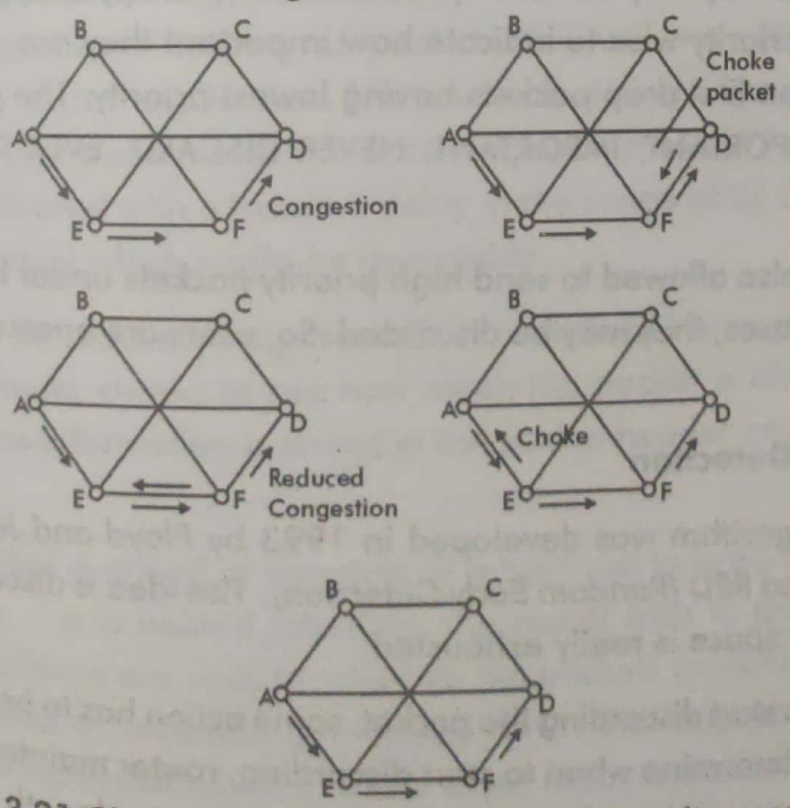

Figure 3.21 Choke packet that affects each hop it passes through

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If there is heavy congestion from A to D, destination sends choke packet back to previous hop F. F is required to reduce the flow to D. F has to allocate more buffer, because the source A is still sending packets heavily. Choke packets reaches F, which tells E to reduce the flow. Finally, the choke packets reaches A and the flow of packets slows down. Congestion can be brought down without losing any packets.

3.3.5 Load Shedding

The load shedding is the ultimate method which can applied if any of those previous approaches did not work. If the router is accumulated with packets that they cannot handle, they just throw them away or ignore. All packets can be thrown away. Because discarding of packets depend upon the type of applications running. For example, in case of file transfer, old packet is worth more than a new one. Because dropping of packets will cause a gap at receiver that may force packets for retransmission. In case of multimedia application, a new packet is more important than an old one.

Intelligent discard policy can be implemented. In this applications must mark their packets in priority wise to indicate how important they are. Based on the priority, router can first drop packets having lowest priority. The priority levels can be VERY IMPORTANT, IMPORTANT, NEVER DISCARD, EVER DISCARD and DISCARD.

Senders are also allowed to send high priority packets under light load. But as the load increases, they may be discarded. So, users are encouraged to stop sending them.

Random Early Detection

A popular algorithm was developed in 1993 by *Floyd and Jacobson*. This algorithm is called *RED (Random Early Detection)*. The idea is discarding packet before all buffer space is really exhausted.

When routers start discarding the packet, some action has to be taken before it is too late. To determine when to start discarding, router maintains a running average of their queue lengths. When the average queue length on some line

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exceeds a threshold, the line is said to be congested and some action must be taken.

Router can detect, which source is causing a problem or trouble.

How router tell the source about the problem?

i) *By send it a choke packet*

ii) *Just discard the selected packet and donot report it. Source will not get acknowledgement for these discarded packets and take action. Source comes to know that lost packets are generally caused by congestion. Hence source responds by slowing down the transmission.*

iii) *The above approach can not be applied for wireless, because most losses are due to noise on the air link.*

3.3.6 Jitter Control

The variation in the packet arrival time is *jitter*. In some applications like audio and video *streaming*, packets are delivered at a *constant transmit time*. If there is variation on delivery of packets, this will have jitter effect on the performance of any application. The unevenly arrival of packets have impact on quality of sound and movie. For streaming applications, 99 percent of packets has to be delivered with a transmit delay in the range of 24.5 m sec to 25.5 m sec (for example) which might be acceptable.

Jitter can be reduced long the path at each hop. When packet arrives at a router, the router checks to see how much the packet is ahead or behind its schedule. This information is stored in the packet header and updated at each hop.

If the packet is ahead of schedule, it is held just long enough to get it back on schedule. If it is behind schedule, the router tries to get it out of its line quickly. Algorithms are used to schedule the packets on outgoing line. Packets that are ahead of schedule get slowed down and packets that are behind schedule gets speeded up. Both these cases reduces the amount of jitter.

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In some applications like video on demand, jilter can be eliminated by buffering the receiver and fetching data for reply from the buffer, instead of the network in real time. In case of internet, telephony and video conferencing applications, buffering concept is not acceptable.

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